

Section 2.2: The Reading Order Problem

In PDF text extraction, reading order refers to reconstructing text in the logical sequence a human reader naturally follows (e.g., top-to-bottom, left-to-right within columns).

When a system naively dumps text streams directly from a PDF into a Language Model or ATS database, the extracted sequence often mangles two-column resumes, sidebars, and floating text boxes into unreadable, interleaved text.

1. Why PDF Object Stream Order Fails (Visual vs. Logical/Stream Order)

PDFs are vector graphic instruction sets, not structured text documents. Characters and text blocks are placed inside BT...ET stream operators using absolute spatial coordinates, rendered in the order they were written by the PDF generator.

- **Z-Index / Creation Order vs. Reading Order:** A designer creating a resume in Figma or Canva might write the main body text first, add the right column, and finally draw a left sidebar. The resulting PDF stream stores the text objects in that exact creation sequence.
- **The Naïve Extraction Traversal:** If a parser traverses the PDF graph without spatial analysis, it pulls text in byte-stream order.

Example of Interleaved Extraction Failure

Given a two-column resume:

LEFT COLUMN (Sidebar)	RIGHT COLUMN (Main Content)
-----	-----
Skills	Work Experience
Python	Senior Engineer at Company A
Java	Built scalable microservices
AWS	Led a team of 5 engineers

A naïve stream extraction or unguided horizontal scan produces:

"Skills Work Experience Python Senior Engineer at Company A Java Built scalable microservices AWS Led a team of 5 engineers"

The semantic connection between job titles, descriptions, and skill sets is completely destroyed.

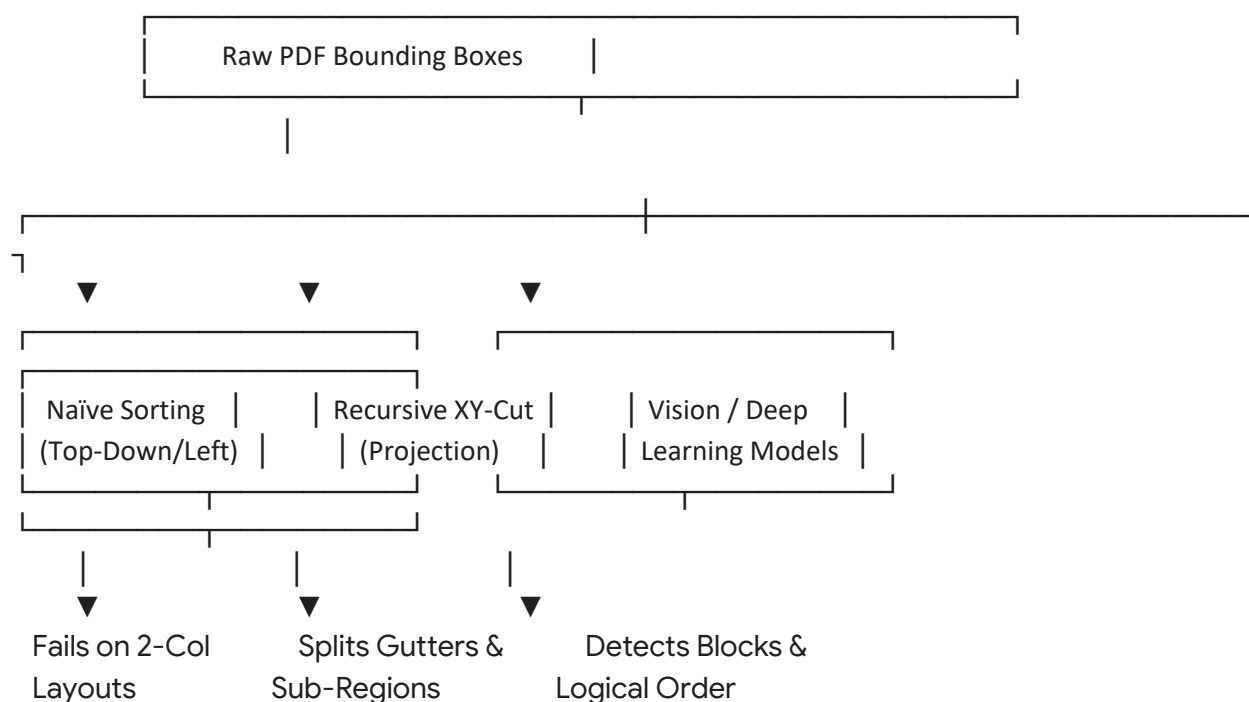
2. Multi-Column Layout Patterns & Spatial Anomalies

Document layouts in resumes and technical documents present specific layout challenges:

- **Symmetric Two-Column / Three-Column Layouts:** Pages split evenly into two or three vertical channels.
- **Asymmetric Sidebar Layouts (20/80 or 30/70 Split):** A narrow left or right sidebar containing skills, contact info, and education, running alongside a wide main column for experience.
- **Spanning Elements (Full-Width Headers/Footers):** A full-width header (e.g., Candidate Name & Contact Info) at the top, followed by a two-column split, followed by a full-width section banner at the bottom.
- **Floating & Overlapping Text Boxes:** Absolute-positioned text boxes created in desktop publishing software (e.g., InDesign) anchored independently of line height or column flow.
- **Staggered Line Baselines:** The left column uses a font size of 10pt with 12pt line spacing, while the right column uses 11pt font with 14pt line spacing. Their line baselines do not align horizontally across the page.

3. Algorithmic Approaches to Reconstructing Reading Order

Engineers use three primary paradigms to solve reading order:



A. Naïve Spatial Sorting (Top-Down, Left-to-Right)

- **Mechanism:** Sorts all bounding boxes primarily by vertical position y_0 (descending in PDF space), then by horizontal position x_0 (ascending).
- **Why it Fails:** On a two-column page, a line in the left column at $y = 500$ and a line in the right column at $y = 498$ are sorted sequentially. The output alternates between columns

on every single line.

B. Recursive XY-Cut Algorithm (Projection Profiles)

- **Mechanism:**
 - Project all word bounding box densities onto the horizontal axis (X-histogram) and vertical axis (Y-histogram).
 - Locate empty space channels (gutters or whitespace valleys) where pixel/box density falls to zero.
 - Divide the page horizontally or vertically at these gutters into child bounding sub-regions.
 - Recursively apply the cut to child regions until atomic text blocks remain.
 - Traverse the resulting tree to establish logical reading order.
- **Strength:** Highly deterministic, lightweight, fast, and does not require GPU acceleration.
- **Weakness:** Fails when columns overlap slightly, when background graphics obscure gutters, or when text is aligned diagonally.

C. Delaunay Triangulation & Spatial Graph Networks

- **Mechanism:** Treats each word/character bounding box as a node in a spatial graph. Constructs a Delaunay triangulation network between nearby bounding boxes and evaluates edge features (angle, distance, font similarity, line alignment) to cluster words into reading chains.

D. Vision & Deep Learning Layout Detection (LayoutLM, Marker, Document AI)

- **Mechanism:** Renders the PDF page into a raster image and passes it through an object detection / visual transformer model (e.g., LayoutLMv3, Marker, or YOLO-Doc).
- **Strength:** Models recognize section headings, sidebars, tables, and figures visually regardless of underlying object stream bugs.
- **Weakness:** Higher computational latency and monetary cost compared to pure deterministic algorithms.

4. Tool Behaviors & Reading Order Capabilities

Different libraries handle reading order using distinct strategies:

Tool / Engine	Reading Order Mechanism	Multi-Column Handling
PyMuPDF (fitz)	Exposes sort=True in get_text("blocks") using spatial heuristics.	Good on clean layouts; can misorder staggered multi-column baselines.

pdfplumber	Extracts raw character/word coordinates (top, x0).	Requires custom Python heuristics (e.g., column bounding-box filtering).
Marker	Combines layout detection heuristics with ML models.	Excellent; reconstructs multi-column content directly into ordered Markdown.
Adobe PDF Extract / AWS Textract	Deep learning visual layout parsers returning reading-order JSON trees.	High precision across asymmetric sidebars, callout boxes, and complex multi-column documents.

5. Edge Cases & Failure Modes in Production

- **Gutter Bleed:** When column gaps are narrow (e.g., < 10 pt), horizontal projection algorithms fail to detect the separation, causing words across columns to merge.
- **Asymmetric Column Termination:** Left column ends halfway down the page while the right column continues to the bottom. Parsers must accurately determine when to merge back into a single full-width layout channel.
- **Mid-Page Layout Transitions:** Pages that switch from single-column (Profile Summary) to two-column (Experience & Education) and back to single-column (Certifications).